📊 Telecom Customer Churn Prediction Report

Client: No-Churn Telecom

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1. Executive Summary

No-Churn Telecom, a leading telecom provider in Europe, is facing a churn rate exceeding 10% due to increasing competition. Traditional retention strategies have not been effective, necessitating a data-driven approach.

This project leverages **Machine Learning (ML)** to:

- Identify customers likely to churn
- Analyze key drivers behind churn
- Generate **churn risk scores**
- Enable targeted retention strategies

The final solution uses a **Random Forest model** to predict churn and provide actionable business insights.

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2. Business Problem

- High customer churn (>10%)
- Ineffective blanket retention campaigns
- Lack of proactive churn identification

Objective

Develop a predictive model to:

- Identify high-risk customers
- Reduce churn through targeted interventions
- Improve customer retention and profitability

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3. Data Overview

Source

- MySQL Database: project_telecom
- Table: telecom_churn_data
- Records: **243,553 rows**
- Features: **14 columns**

Key Features

- Demographics: age, gender, state, city
- Usage: calls, SMS, data usage
- Financial: estimated salary
- Target: **Churn (0 = No, 1 = Yes)**

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4. Data Preprocessing

Steps Performed

- Removed irrelevant columns:
    - customer_id, pincode, date_of_registration
- Converted data types appropriately
- No missing values or duplicates found

Feature Engineering

Created new variables:

- total_calls = calls_made + sms_sent
- total_charge = calls_made + sms_sent
- total_mins = calls_made + data_used

These improved model performance by capturing customer behavior patterns.

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5. Exploratory Data Analysis (EDA)

Key Observations

- Churn rate ≈ **20%**

- Certain telecom partners show higher churn
- High data usage correlates with churn
- Customer behavior varies significantly across regions

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6. Model Development

Algorithms Used

1. Logistic Regression
2. K-Nearest Neighbors (KNN)
3. Random Forest (Final Model)

Train-Test Split

- Training: 70%
- Testing: 30%
- Stratified sampling used

Feature Scaling

Applied StandardScaler for distance-based models

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7. Model Evaluation

1. Logistic Regression

- Accuracy: **~80%**
- Failed to predict churn class (recall = 0)

2. KNN

- Accuracy: **~76%**
- Poor churn detection (low recall)

3. Random Forest (Final Model)

- Accuracy: **~80%**
- Captures non-linear relationships

- Provides feature importance

⚠️ **Important Insight:**
All models struggle with **class imbalance**, leading to poor churn detection (recall for churn = 0 in RF).

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8. Feature Importance (Key Drivers of Churn)

Top influencing features:

1. Estimated Salary
2. Data Usage
3. Total Minutes
4. Calls Made
5. Age
6. SMS Sent
7. Total Charges
8. State & City

Insight

- High usage customers are more likely to churn
- Behavioral features matter more than demographics

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9. Churn Risk Scoring

Approach

- Used predict_proba() from Random Forest
- Generated:
    - **Churn Risk Score (0–1)**
    - **CHURN_FLAG (1 if ≥ 0.5)**

Business Value

- Enables **proactive retention**
- Helps prioritize high-risk customers

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10. Key Business Insights

- High usage customers tend to churn more
- Customer dissatisfaction is reflected in usage patterns
- Not all customers are equally valuable or risky
- Regional and behavioral differences impact churn

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11. Recommendations

1. Target High-Risk Customers

- Use CHURN_FLAG for proactive retention campaigns

2. Improve Customer Service

- Focus on customers with high engagement
- Reduce resolution time

3. Personalized Offers

- Avoid generic discounts
- Tailor plans based on usage patterns

4. Monitor Behavior

- Track unusual changes in usage

5. Deploy Model

- Integrate into CRM systems

6. Retrain Model

- Quarterly updates to maintain accuracy

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12. Limitations

- Class imbalance affects churn prediction
- Some unrealistic values (negative usage) present
- No time-series behavior modeling
- Limited feature diversity

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13. Future Improvements

- Apply **SMOTE / class balancing**
- Try advanced models:
    - XGBoost / LightGBM
- Add customer service interaction data
- Use time-based features (trend analysis)
- Build real-time prediction pipeline

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14. Conclusion

This project successfully demonstrates how machine learning can be used to predict telecom churn and generate actionable insights.

The **Random Forest model** was selected as the final solution due to:

- Strong performance
- Interpretability
- Business relevance

The model enables **data-driven decision making**, helping No-Churn Telecom reduce churn and improve customer retention.